



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO M.A.M. AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

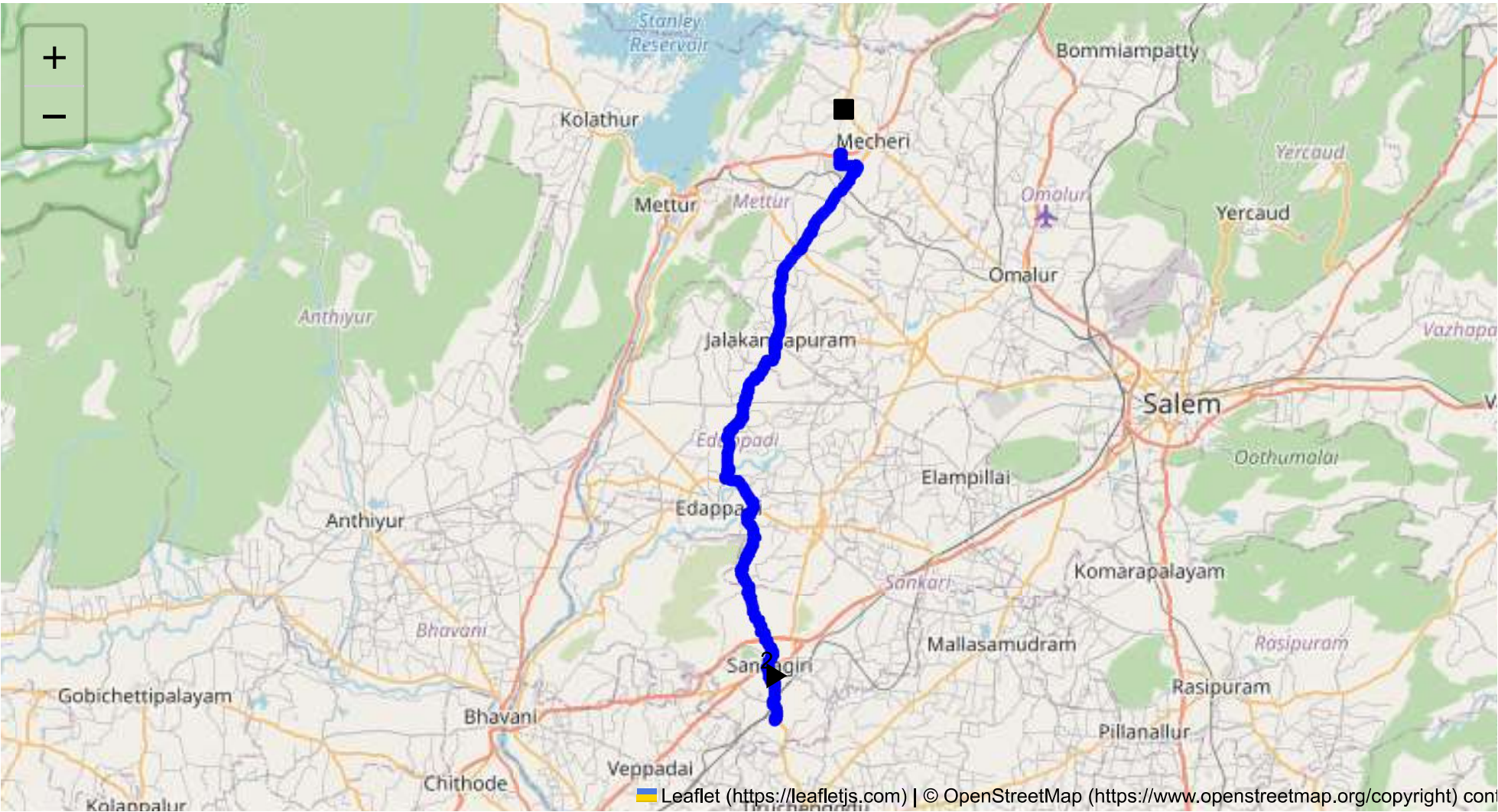
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 51.65 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (11.825472, 77.920242)



Welcome to the Journey Risk Management Study

Certainly, here is an analysis of the route from Sangagiri to Mecheri:

1. Overview of the Route Map

The route from Sangagiri to Mecheri is primarily along NH44, taking a northern direction through rural and semi-urban regions. This route spans approximately 51.65 kilometers with a typical travel time of about 1

hour and 38 minutes for heavy vehicles transporting hazardous materials.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with temperatures high throughout the year. The main weather-related hazards:

- **Monsoon Season (June to September):** Heavy rains can cause waterlogging and reduce visibility.
- **Summer (March to May):** Extreme heat, which can cause road surfaces to become sticky and slippery.
- **Cyclones/Depressions:** Although rare inland, they can still bring heavy rains and winds, particularly in late-year months.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, peak traffic occurs in the morning from 8:00 AM to 10:00 AM and in the evening from 5:00 PM to 8:00 PM when local commuters are most active.
- **Congestion-Prone Areas:** Urban areas close to intersections and markets in towns along the way such as Tharamangalam are likely locations for congestion.

4. Assessment of Road Quality and Infrastructure

- **NH44:** Generally well-maintained with adequate signage and lane markings. The road quality tends to be better on highways with occasional potholes and uneven surfaces closer to rural exits.
- **Bridges/Culverts:** Frequent on this route; ensure compliance with weight limits.

5. Suggestions for Alternative Routes for Emergencies

- **Via Tharamangalam – Danishpet:** This alternative route via State Highway, though longer, can be used in the event of roadblocks or severe weather disturbances on NH44.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Hazardous materials require specific permits.
- Restrictions generally apply during peak seasonal festivals due to increased pedestrian traffic.
- Night-time travel may be prohibited or require additional permits.

7. Overview of Historical Incidents

- Historical data suggests occasional heavy vehicle accidents during monsoon due to skidding on wet roads.
- Instances of overturned trucks near narrow bends and steep gradients leading to hazardous spills.

8. Environmental Considerations and Sensitive Areas

- Sections travel close to forested regions; adherence to designated transport routes to avoid disturbing wildlife is crucial.
- Villages along the route request reduced noise pollution measures.

9. Analysis of Communication Coverage

- **Strong Coverage:** Near urban areas and along major highways.
- **Potential Dead Zones:** Rural stretches and valleys where signal might drop.

10. Estimated Emergency Response Times

- **Urban Areas:** Approx. 20-30 minutes.
- **Rural Areas:** Approximately 40 minutes to 1 hour, depending on proximity to local emergency facilities.

12. Overall Summary of Risk Assessment

The route has generally good infrastructure but poses potential risks primarily related to weather conditions and rural stretches where communication and emergency response could be delayed. Transporters should be equipped with emergency kits, communication devices, and detailed understanding of local traffic regulations and historical considerations while planning the voyage. Regular updates from local authorities and communication with control centers are vital to ensure safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45000, 77.87213	30 KM/Hr
7	Turn	High	11.45028, 77.87202	15 KM/Hr
8	Turn	Medium	11.45309, 77.87325	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.47451, 77.86914	15 KM/Hr
11	Turn	High	11.47920, 77.86989	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.48070, 77.87166	30 KM/Hr
13	Turn	Medium	11.48155, 77.87174	30 KM/Hr
14	Turn	Medium	11.48277, 77.87250	30 KM/Hr
15	Turn	Medium	11.49496, 77.86660	30 KM/Hr
16	Turn	Medium	11.52571, 77.85522	30 KM/Hr
17	Turn	Medium	11.53598, 77.85142	30 KM/Hr
18	Turn	Medium	11.53857, 77.85087	30 KM/Hr
19	Turn	Medium	11.55027, 77.85506	30 KM/Hr
20	Turn	Medium	11.55460, 77.85624	30 KM/Hr
21	Turn	High	11.57893, 77.85430	15 KM/Hr
22	Turn	Medium	11.57903, 77.85441	30 KM/Hr
23	Turn	Medium	11.60960, 77.84159	30 KM/Hr
24	Turn	Medium	11.61795, 77.84020	30 KM/Hr
25	Turn	Medium	11.62066, 77.84105	30 KM/Hr
26	Turn	Medium	11.64055, 77.84600	30 KM/Hr
27	Turn	Medium	11.64173, 77.84878	30 KM/Hr
28	Turn	Medium	11.64522, 77.85113	30 KM/Hr
29	Turn	Medium	11.66451, 77.85595	30 KM/Hr
30	Turn	Medium	11.67322, 77.86045	30 KM/Hr
31	Turn	Medium	11.67368, 77.86223	30 KM/Hr
32	Turn	Medium	11.68196, 77.86690	30 KM/Hr
33	Turn	High	11.68488, 77.86889	15 KM/Hr
34	Turn	Medium	11.68463, 77.87262	30 KM/Hr
35	Turn	Medium	11.68486, 77.87288	30 KM/Hr
36	Turn	Medium	11.69289, 77.87444	30 KM/Hr
37	Blind Spot	Blind Spot	11.69469, 77.87589	10 KM/Hr
38	Turn	High	11.69491, 77.87546	15 KM/Hr
39	Turn	High	11.69501, 77.87546	15 KM/Hr
40	Turn	High	11.69551, 77.87442	15 KM/Hr
41	Turn	Medium	11.69908, 77.87472	30 KM/Hr
42	Turn	Medium	11.69934, 77.87500	30 KM/Hr
43	Turn	Medium	11.76019, 77.89047	30 KM/Hr
44	Turn	Medium	11.75964, 77.89200	30 KM/Hr
45	Blind Spot	Blind Spot	11.75947, 77.89214	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
46	Blind Spot	Blind Spot	11.81655, 77.93178	10 KM/Hr
47	Turn	Medium	11.81736, 77.93039	30 KM/Hr
48	Turn	Medium	11.81715, 77.92168	30 KM/Hr
49	Turn	Medium	11.81817, 77.91984	30 KM/Hr
50	Turn	High	11.81986, 77.91933	15 KM/Hr
51	Turn	Medium	11.82031, 77.91975	30 KM/Hr
52	Turn	High	11.82264, 77.92000	15 KM/Hr
53	Turn	High	11.82281, 77.91941	15 KM/Hr
54	Turn	High	11.82570, 77.91960	15 KM/Hr
55	Turn	High	11.82587, 77.92015	15 KM/Hr
56	Turn	Medium	11.82576, 77.92019	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
7	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
9	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
10	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium